



DUMINDU EKANAYAKA

GRADUATED ENGINEER

MECHATRONICS and ROBOTICS Engineering



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Bulugahapitiya, Eheliyagoda.



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2001 - 04 - 21

EDUCATION

BEng (Hons) Mechatronics and Robotics Engineering (Top-Up)
- Upper Second Class Honours
Anglia Ruskin University (UK)
Sep 2025 – Sep 2026

Pearson BTEC Higher National Diploma (HND) in Mechanical Engineering - Merit Pass
CINEC CAMPUS
Sep 2023 – Sep 2025

ENGINEERING PROJECTS

Self-Balancing Mobile Robot

- Developed an ESP32-based self-balancing robot using MPU6050 IMU sensing and PID control.
- Integrated encoder feedback, wireless web control and obstacle detection for improved control and functionality.
- Performed testing and PID tuning to improve stability, disturbance recovery and movement performance.
- Designed the mechanical chassis and supporting engineering components.

IoT-Based Health Monitoring System

- Developed an ESP32-based IoT monitoring system using MAX30102, DS18B20 and DHT11 sensors for heart rate, SpO₂, temperature and humidity.
- Implemented Blynk IoT wireless monitoring, LCD display integration and embedded sensor processing.
- Designed the system enclosure using SOLIDWORKS.

ABOUT ME

Mechatronics and Robotics Engineering graduate with industrial training experience in mechanical maintenance, production systems and engineering problem-solving. Hands-on experience in robotics, embedded systems, automation and control using ESP32, Arduino, sensors, PID control and PLC/HMI systems. Skilled in SOLIDWORKS, C++, Python and MATLAB. Seeking a Graduate Engineer role in mechatronics, robotics or automation.

EXPERIENCE

Engineer Intern — Rocell (Eheliyagoda) Mar 2025 – Sep 2025 | On-site

- Supported mechanical and factory maintenance in a production environment, including conveyors, electric motors, hydraulics, pumps and gearboxes.
- Assisted with production-stoppage analysis, preventive/corrective maintenance and process improvement activities.
- Developed 3D machine and engineering drawings using SOLIDWORKS.
- Contributed to a pallet-loading efficiency improvement initiative while following industrial safety practices.

TECHNICAL SKILLS

- Robotics & Embedded:** Robotics, Embedded Systems, ESP32, Arduino, Sensor Integration, IoT
- Automation & Control:** Industrial Automation, PLC, HMI, PID Control, Motor Control, VFD
- Mechanical:** Mechanical Maintenance, Electric Motors, Hydraulics, Pumps, Gearboxes, Conveyors
- Software & Electronics:** SOLIDWORKS, MATLAB, Arduino IDE, C++, Python, PCB Design, Circuit Design

CERTIFICATIONS

- Robotics** — Southern IRAA Pvt Ltd
- Artificial Intelligence** — Southern IRAA Pvt Ltd
- Industrial Automation Engineering** — EPIC Campus (PLC & HMI Programming)
- Diploma in English**

LANGUAGES Sinhala – Native | English – Intermediate/Advanced

REFERENCES

B.J.M.Cooray
Training Engineer
CINEC Campus
IT Park
Malabe

Hasitha Wipulaguna B.Sc Eng(Hons) ,Msc/PGDip
in Materials Science Lecturer in Mechanical
Engineering, CINEC Campus
[0770843545](tel:0770843545)

CONCLUSION

I hereby certify that the above-mentioned information is true and accurate to the best of my knowledge.

Dumindu Ekanayaka